Vector Spaces: a Compact Lecture

A fully synthetic demonstration page

REVISION LANE

Revision cue closure -> operations stay inside

Definition

A vector space is a collection of objects that can be added and scaled without leaving the collection.

中譯｜向量空間是一個在向量加法與純量乘法下仍保持封閉的集合。

SOURCE ARTIFACT

EXPLANATION LANE

Intuition A vector space is a sandbox closed under two moves.

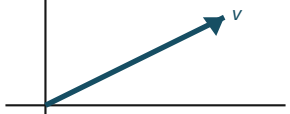
Example

Pairs (x, y) form a vector space: add coordinates componentwise and multiply both coordinates by the same scalar.

$$a(u + v) = au + av$$

中譯｜每個運算都逐座標進行，因此結果仍是一組二維座標。

Figure



中譯｜箭頭同時表達方向與大小。

Figure 1. A vector as directed displacement

Figure guide Read tail, head, direction, then length.

Common mistake nonzero does not mean independent

Result

Two non-parallel vectors in the plane provide a basis for every planar vector.

中譯｜平面中兩個不平行向量可唯一表示任意平面向量。

Proof

Place the two vectors as columns of a 2 by 2 matrix. Non-parallel columns give a nonzero determinant, so the coordinate system has a unique solution.

$$det[u\ \ v] != 0$$

中譯｜行列式非零等價於兩欄獨立，因此座標解存在且唯一。

Exercise

Checklist 1. determinant 2. solve coordinates 3. interpret

Decide whether u = (1, 1) and v = (2, -1) form a basis, then express w = (5, 1) in that basis.

Derivation island - exercise solution

det[[1, 2], [1, -1]] = -3, so the vectors form a basis. Solving c1 u + c2 v = w gives c1 = 7/3 and c2 = 4/3.